KILLESTER COLLEGE

STRENGTH & KINDLINESS

FURTHER MATHEMATICS UNIT 3
APPLICATION TASK SAC 1 2020

CORE – DATA ANALYSIS : CORONA VIRUS

NAME: ___________________________________ TEACHER: ________________

Dates: PART ONE - Week 6 Term 2 Remotely (all students) Monday 18th May 9:00 – 10:40am
PART TWO - Term 3 (all students)

Duration: PART ONE – 10 minutes administration and roll marking, 10 minutes reading time plus 80 minutes
PART TWO – 10 minutes reading time plus 110 minutes

Materials allowed: 1 x Bound reference, CAS calculator, scientific calculator, ruler, pens/pencils

Outline of SAC

At the end of each session the student must:
1. Submit their SAC,
2. Sign to state that they have submitted all these items and that the work is their own.

Total : 67 marks

PART ONE : 26 marks

PART TWO : 41 marks

STUDENT DECLARATION

- I declare that the work completed in this assessment is my own work and demonstrates my own abilities and knowledge and does not involve plagiarism or collaboration with other students
- I declare that I have not knowingly assisted another student in a breach of rules
- I understand that I must not cheat or assist other students to cheat, including taking any action that gives or attempts to give me or another student an unfair advantage
- I declare that I have not engaged in unacceptable forms of assistance including; use of, or copying, another person's work or other resources without acknowledgement; corrections or improvements made or dictated by another person
- I declare that I have not contracted another person to do the work for me or allowed another person to edit and substantially change my work.
- So that the teacher can properly assess my work, I give my teacher permission to reproduce this work and provide a copy to another member of staff for the purpose of cross checking and moderation and to take steps to authenticate its originality.
- I understand that if a teacher is in doubt about the authenticity of my work, an authentication panel will conduct an investigation into any breach of VCAA rules
- I understand that any School-assessed Coursework marks given to me are conditional and may be altered if: the work is atypical of other work produced by me; is inconsistent with the teacher's knowledge of my ability; the work contains unacknowledged material; the work as not been sighted and monitored by the teacher during it development (if relevant)
- I understand that total scores for School-assessed Coursework may change as a result of statistical moderation

Signed: ___________________________________

Date: _____________________________________

In order to satisfactorily complete this SAC students must demonstrate an understanding of the following three outcomes. These outcomes are covered throughout the SAC.
The following table summarises what each outcome entails in terms of key knowledge and skills. Please note that not all points are covered in this SAC- indicated by an NA (not assessed).

Outcome 1 (10 marks): On completion of this unit the student should be able to define and explain key concepts and apply related mathematical techniques and models as specified in Area of Study 1 in routine contexts. To achieve this outcome the student will draw on knowledge and skills outlined in Area of Study 1.	
Key knowledge	Satisfactory
• types of data: categorical (nominal and ordinal) and numerical (discrete and continuous)	
• frequency tables, bar charts including segmented barcharts, histograms, stem plots, dot plots, and their application in the context of displaying and describing distributions	
• log (base 10) scales, and their purpose and application	
• five-number summary and boxplots (including the designation and display of possible outliers)	
• mean and standard deviation	
• normal model and the 68–95–99.7% rule, and standardised values (z-scores)	
• response and explanatory variables	
• two-way frequency tables, segmented bar charts, back-to-back stem plots, parallel boxplots, and scatterplots, and their application in the context of identifying and describing associations	
• correlation coefficient, r, its interpretation, the issue of correlation and cause and effect	
• least squares line and its use in modelling linear associations	
• data transformation and its purpose	
• time series data and its analysis	
Key skills	
• construct frequency tables and bar charts and use them to describe and interpret the distributions of categorical variables	
• answer statistical questions that require a knowledge of the distribution/s of one or more categorical variables	
• construct stem and dot plots, boxplots, histograms and appropriate summary statistics and use them to describe and interpret the distributions of numerical variables	
• answer statistical questions that require a knowledge of the distribution/s of one or more numerical variables	
• solve problems using the z-scores and the 68–95–99.7% rule	
• construct two-way tables and use them to identify and describe associations between two categorical variables	NA
• construct parallel boxplots and use them to identify and describe associations between a numerical variable and a categorical variable	
• construct scatterplots and use them to identify and describe associations between two numerical variables	
• calculate the correlation coefficient, r, and interpret it in the context of the data	
• answer statistical questions that require a knowledge of the associations between pairs of variables	
• determine the equation of the least squares line giving the coefficients correct to a required number of decimal places or significant figures as specified	
• distinguish between correlation and causation	
• use the least squares line of best fit to model and analyse the linear association between two numerical variables and interpret the model in the context of the association being modelled	
• calculate the coefficient of determination, r^2 , and interpret in the context of the association being modelled and use the model to make predictions, being aware of the problem of extrapolation	
• construct a residual analysis to test the assumption of linearity and, in the case of clear non-linearity, transform the data to achieve linearity and repeat the modelling process using the transformed data	
• identify key qualitative features of a time series plot including trend (using smoothing if necessary), seasonality, irregular fluctuations and outliers, and interpret these in the context of the data	
• calculate, interpret and apply seasonal indices	NA
• model linear trends using the least squares line of best fit, interpret the model in the context of the trend being modelled, use the model to make forecasts being aware of the limitations of extending	

forecasts too far into the future	
Outcome 2 (20 marks): On completion of this unit the student should be able to select and apply the mathematical concepts, models and techniques as specified in Area of Study 1 in a range of contexts of increasing complexity. To achieve this outcome the student will draw on knowledge and skills outlined in Area of Study 1	
Key knowledge	
• the facts, concepts and techniques associated with data analysis	
• standard models studied in data analysis and their area of application	
• general formulation of the concepts, techniques and models studied in data analysis	
• assumptions and conditions underlying the use of the concepts, techniques, and models associated with data analysis	
Key skills	
• identify, recall and select facts, concepts, models and techniques needed to investigate and analyse statistical features of a data set with several variables that can include time series data	
• interpret and report the results of a statistical investigation or of completing a modelling or problem-solving task in terms of the context under consideration, including discussing the assumptions in application of these models	
Outcome 3 (10 marks): On completion of this unit the student should be able to select and appropriately use numerical, graphical, symbolic and statistical functionalities of technology to develop mathematical ideas, produce results and carry out analysis in situations requiring problem-solving, modelling or investigative techniques or approaches. To achieve this outcome the student will draw on knowledge and related skills outlined in Area of Study 1.	
Key knowledge	
• the difference between exact numerical and approximate numerical answers when using technology to perform • computation, and rounding to a given number of decimal places or significant figures	
• domain and range requirements for specification of graphs of models and relations, when using technology	
• the relation between numerical, graphical and symbolic forms of information about models and equations and • the corresponding features of those functions and equations	
• similarities and differences between formal mathematical expressions and their representation by technology	
• the selection of an appropriate functionality of technology in a variety of mathematical contexts	
Key skills	
• distinguish between exact and approximate presentations of mathematical results produced by technology, and interpret these results to a specified degree of accuracy in terms of a given number of decimal places or significant figures	
• use technology to carry out numerical, graphical and symbolic computation as applicable	
• produce tables of values, families of graphs and collections of other results using technology, which support general analysis in problem-solving, investigative and modelling contexts	
• identify the relation between numerical, graphical and symbolic forms of information about models and equations and the corresponding features of those models and equations	
• select an appropriate functionality of technology in a variety of mathematical contexts, related to data analysis, and provide a rationale for these selections	
• relate the results from a particular technology application to the nature of a particular mathematical task (investigative, problem solving or modelling) and verify these results	
• specify the process used to develop a solution to a problem using technology, and communicate the key stages of mathematical reasoning (formulation, solution, interpretation) used in this process	

Overall result:	Outcome 1	Outcome 2	Outcome 3

After SACs have been marked, teachers will decide if a student has demonstrated enough knowledge throughout their SAC to be given a satisfactory. If a student can’t demonstrate through her SAC that she understands the content covered then the student must demonstrate her knowledge by either: 1. Showing her teacher her up-to-date workbook; and/or, 2. Completing a short test.

SAC

INTRODUCTION:

The Corona Virus, otherwise known as Covid-19 became a pandemic in 2020. This SAC includes a hypothetical case of a fictional scientist (Professor Chrissy Higgins) investigating the effectiveness of an anti-viral drug in combatting the corona virus. To measure the level of infection, a blood sample is taken and a molecular test is performed. The molecular test measures the average number of molecular markers per ml in the blood. These molecular markers (*markers*) are a good measure for active infection of the virus.

In order to perform tests quickly, and before trialling this drug on humans, Professor Higgins must first look at the effect this drug has on laboratory rats. This allows large numbers to be sampled to achieve a statistical analysis. Rats showing symptoms of Corona virus typically have an average number of molecular markers (*markers*) per ml greater than 800.

Professor Higgins wishes to investigate:-

- the effectiveness of different doses of the drug
- the side effects of the drug on the health of the rats
- the possibility of predicting how infectious the virus is after a period of time

This SAC also investigates the current data of the actual number of Corona virus cases per country as at the 12th of March 2020.

PART ONE

Comparing dosages. (28 marks)

Professor Higgins wants to explore how much of the drug needs to be given (*dosage*) to decrease the average number of molecular markers per ml (*markers*) in rats.

To test the effect of this drug on the average number of molecular markers per ml (*markers*), she has given different dosages to a sample of 1000 rats and recorded their *markers* after one week. The 1000 rats were all showing symptoms of Corona virus and had an average number of molecular markers (*markers*) per ml of approximately 800.
The following table outlines the number of rats given the different dosages and their *markers*.

***Markers* (*per ml*)**	**No dosage**	**Low dosage**	**Medium dosage**	**High dosage**
0 - < 100	0	0	0	0
100 - < 200	0	0	0	0
200 - < 300	8	1	5	0
300 - < 400	20	19	24	0
400 - < 500	78	61	69	0
500 - < 600	92	72	77	0
600 - < 700	75	58	63	8
700 - < 800	18	11	19	15
800 - < 900	0	2	6	58
900 - < 1000	0	0	0	68
≥ 1000	1	0	0	72

1. Professor Higgins is using bivariate data. What is the difference between univariate data and bivariate data? (1 mark)

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2. What are the variables Professor Williams is looking at? (2 marks)

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3. Which variable is the explanatory variable and which variable is the response variable? Explain your answer. (2 marks)

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4. Are the variables numerical continuous, numerical discrete, categorical ordinal or categorical nominal? (2 marks)

EXPLANATORY VARIABLE ____

RESPONSE VARIABLE ____

5. What is the purpose of not giving a dosage to one group? (1 mark)

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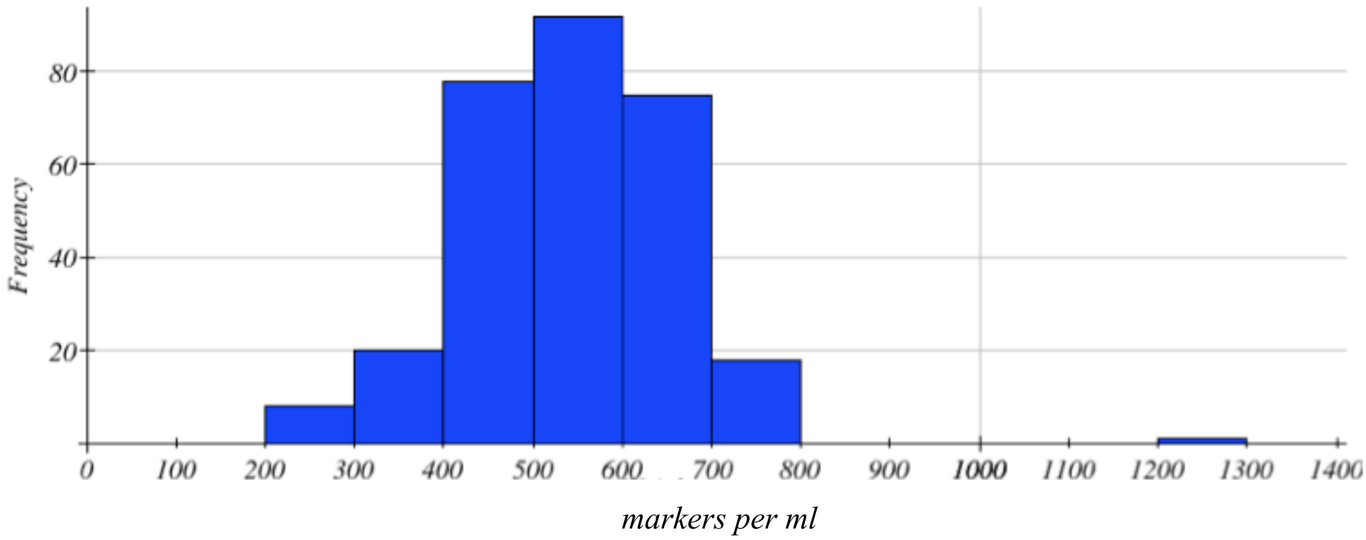
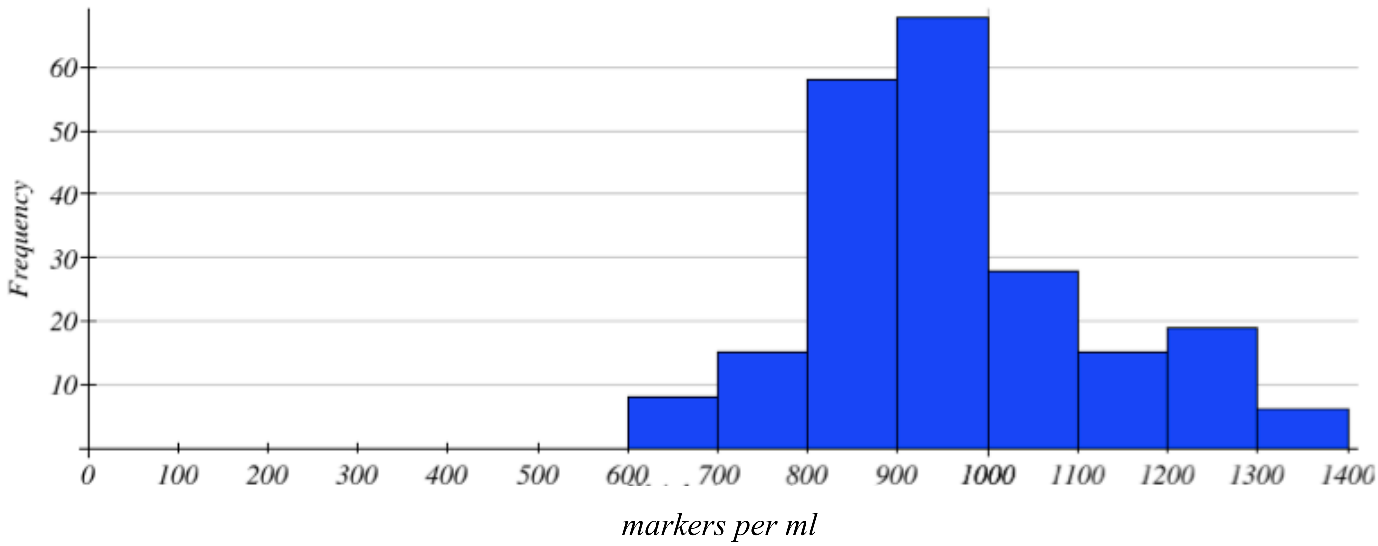
The following histograms illustrate the *markers* of two of the four groups of rats.

FIRST HISTOGRAM



SECOND HISTOGRAM



6. Which two groups do the histograms refer to? (2 marks)

First: ____

Second: ____

7. What is the modal class for each group? (2 marks)

First: ____

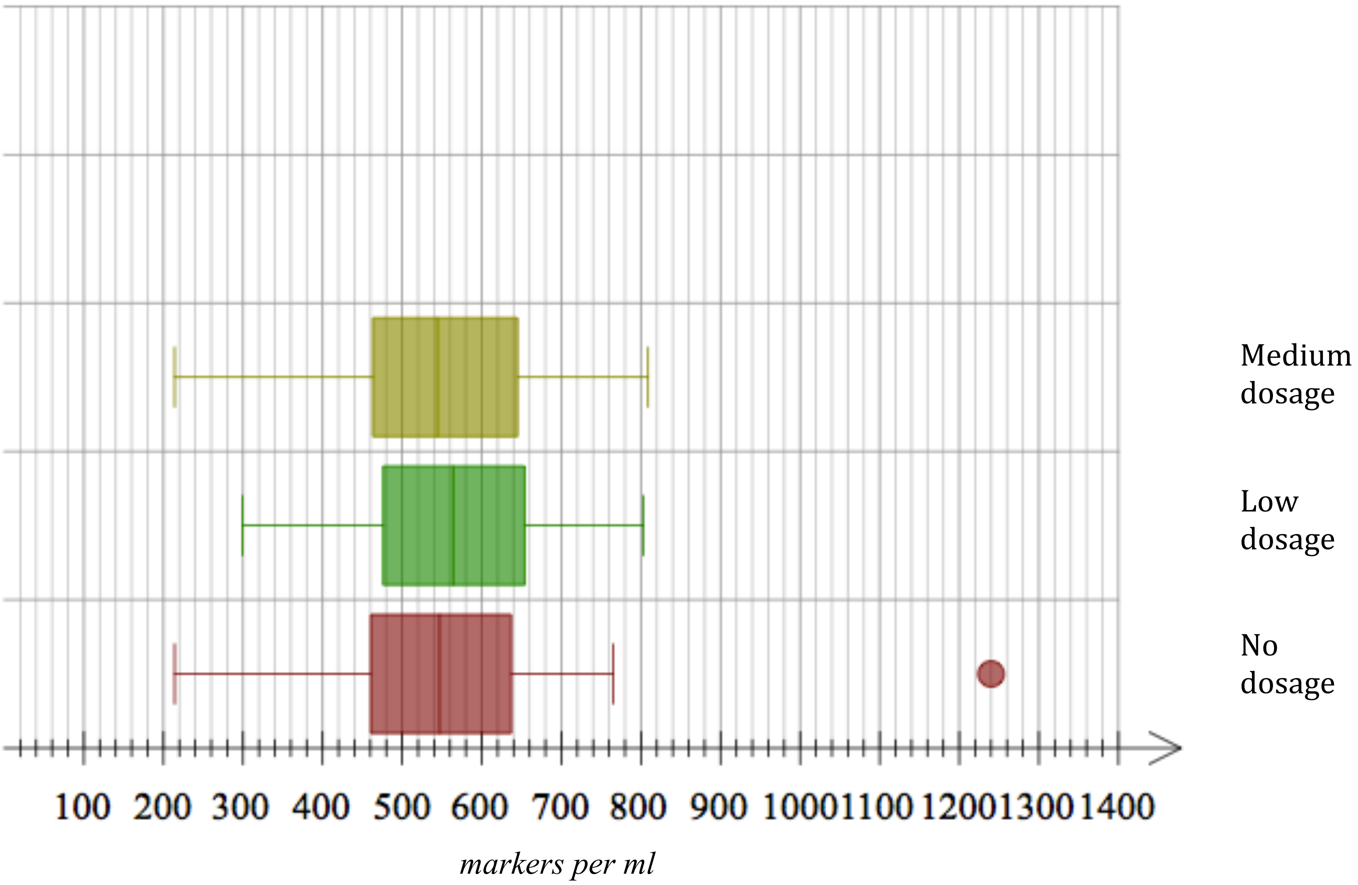
Second: ____

8. Describe the distribution of each histogram. (2 marks)

First: ____

Second: ____

Professor Higgins has developed the following parallel box plots to represent the *markers* for three of the four groups of rats.



9. What do the three box plots drawn above indicate about the relationship between *markers* and drug dosage? (2 marks)

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The box plot for the 'High Dosage' group is missing. Following is the five number summary for the 'High Dosage' group to compare to the box plots on the above grid. Note that there are no outliers for this group.

Min=623

Q_1=784

Med=946

Q_3=1076

Max=1325

10. Approximately, what percentage of 'High dosage' rats have more *markers* than the other three groups of rats? (1 mark)

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11. There is a clear outlier for the 'No Dosage' group. The value of this outlier is 1240 markers.

a. Using the 'No Dosage' box plot, prove that 1240 is an outlier.

(When typing your answer use Q3 rather than Q_3.) (2 marks)

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b. If this outlier was removed, how would it affect the 5 number summary for this group? (1 mark)

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c. How would removing this outlier affect the mean for this group? (1 mark)

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12. Compare the ‘High Dosage’ data with the ‘No Dosage’ data using statistics. (3 marks)

13. What can you conclude about the effect of this drug on the Corona virus in the rats? Explain your reasoning by making reference to the four groups of data. (2 marks)

PART TWO (41 marks)

NAME: ________________________________ **TEACHER:** _______________

Categorising infectiousness.

From previous research, Professor Higgins has found that the *markers* in adult rats with Corona virus, is normally distributed with a mean of 1138 markers and a standard deviation of 98 markers.

1. Draw the bell-shaped curve representing the normal distribution with a mean of 1138 and a standard deviation of 98. Include values three standard deviations from the mean. (2 marks)

Professor Higgins has developed a classification system depending on the *markers* in the rat. The classification system is based on the following:

2. Complete the following table. (2 marks)

Classification	Interpretation	
Extremely contagious	Above one standard deviation above the mean	*markers* ≥ 1236
Contagious		844 ≤ *markers* < 1236
Unlikely to be contagious	Below three standard deviations below the mean	

3. Approximately how many of the 1000 rats fall into the extremely contagious category? (1 mark)

4. Professor Higgins took a sample of 4 rats and calculated their standardised score.

Rat	z-score	Marker classification
A	-3.4	
B	1.3	
C	-0.8	
D	3.1	

a. State the marker classification for each rat by completing the above table. (2 marks)

b. Confirm your classification in question **4a** by calculating the *markers* for Rat A, Rat B, Rat C & Rat D.

(2 marks)

Here is a reminder from PART ONE to help you answer the following questions.

Markers (per ml)	**600 - < 700**	**700 - < 800**	**800 - < 900**	**900 - < 1000**	**≥ 1000**
High Dosage	8	15	58	68	72

This is the five number summary for the 'High Dosage' group. Note that there are no outliers for this group.

Min=623 , Q_1=784, Med=946, Q_3=1076, Max=1325

5. Using the information above:

a. Approximately what percentage of the 'High Dosage' group are expected to have less markers than 784? (1 mark)

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b. Approximately how many rats from this group are expected to be classified as 'Unlikely to be Contagious'? (2 marks)

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c. How does this answer to **5b** above compare to the normal distribution in question **2**? (2 marks)

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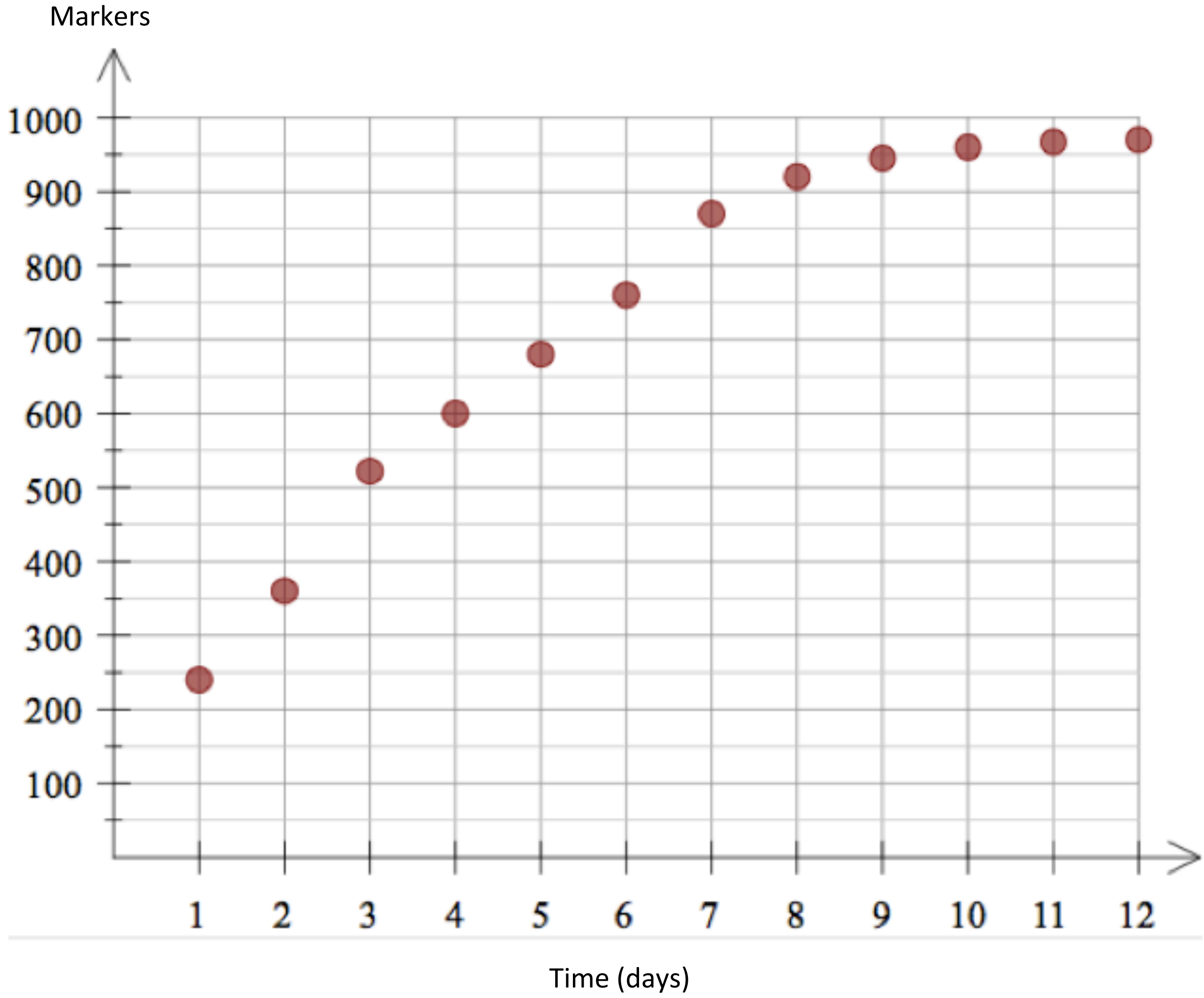
Finding a model.

Professor Higgins has realised that many of the rats given a high dosage of the drug have more markers than the average markers of an infected adult rat. From previous experience, she has noticed that the internal organs of the rat stop working effectively once the rats reach a certain level of the markers in the bloodstream. Professor Higgins needs to develop a model to predict whether the rats given the high dosage will continue gaining markers.

The following table and scatterplot shows the average markers of the rats in the 'High Dosage' group over the period of 12 days.

Time (days)	1	2	3	4	5	6	7	8	9	10	11	12
Markers per ml	240	360	522	600	680	760	870	920	945	960	967	970



6. What is the form of this scatterplot? (1 mark)

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7. What would be the problem with fitting a linear regression model to this data? (1 mark)

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8. Which transformations should be considered in order to linearise this data? (1 mark)

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9. Complete the following table using the transformations you chose in question 8. (3 marks)

Transformation	Coefficient of determination (to 2 decimal places)	Residual plot

10. Which transformation is the best? Explain why. (2 marks)

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11. What is the equation of this transformation model to 3 significant figures and using appropriate variables? (2 marks)

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12. How many markers are the 'High Dosage' rats expected to have after they have been infected for three weeks to the nearest number? (2 marks)

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13. The internal organs of a rat will stop working approximately when the average number of markers is more than 1400.

a. When is it predicted that the 'High Dosage' rats will reach this level to 2 significant figures? (2 marks)

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b. What are the limitations of this prediction? (1 mark)

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Making predictions and the current effect on the world.

Using the High Dosage data on page 16, the transformation equation for is found to be:

$$markers = 180 + 764(\log(days))$$

14. What is the residual for day 8 to 2 decimal places? (1 mark)

15. According to the classification system, rats are considered 'Contagious' if their markers are more than 884.

a. After how many days to 1 decimal place, would you expect the 'High dosage' rats to be considered contagious? (2 marks)

b. What implication does this high dosage of drug have on the quality of life of these rats? (1 mark)

16. The graph on page 16 of the High Dosage data could be considered a time series.

a. What would be the key feature/s of this plot? (1 mark)

b. Show with a clearly marked '**X**' <u>on the graph on page 16</u>, the seven-median smoothed point for day 8. (1 mark)

c. Calculate the six-moving mean for day 8 to 2 decimal places. (2 marks)

Covid-19 Confirmed Cases by Country as at 12/03/20

Country	Total Cases
China	80,796
Italy	12,462
Iran	9,000
S. Korea	7,869
France	2,281
Spain	2,277
Germany	1,966
USA	1,329
Diamond Princess	696
Switzerland	652
Japan	639
Norway	629
Denmark	514
Netherlands	503
Sweden	500
UK	456
Belgium	314
Qatar	262
Austria	246
Bahrain	195
Singapore	178
Malaysia	149
Australia	139
Hong Kong	130
Canada	118
48 countries	10-99
52 countries	1-9
TOTAL = 125	

For this summary the cruise ship Diamond Princess is counted as a country.

17. What percent of these countries have confirmed cases over 1000? (1 mark)

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18. Using the table from the previous page, draw a histogram of the frequency against the log(cases).

(3 marks)

log(cases)	**Frequency**

END OF SAC